Internship project Status W12: CAD Import & smart meshing in Wakis

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Outline

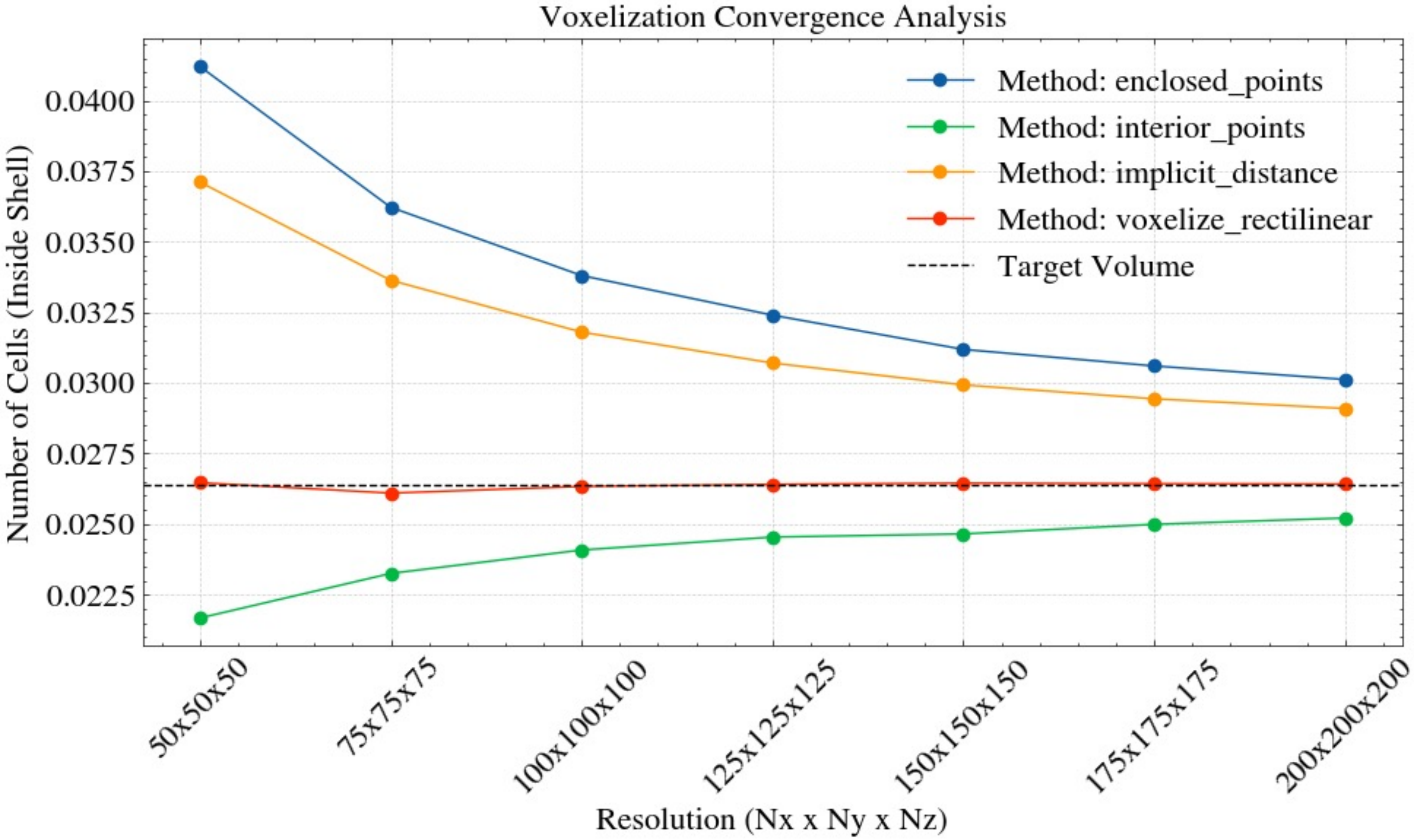
1. **Benchmarks of the methods on the shell & 6L2 device objects mask**
   - Mask volume (quantitative)
     - High grid resolution results
   - Mask slices (qualitative: visual inspection)

2. **Wakis wakefield simulation of the simple shell cavity device for all methods**
   - Voxelize_rectilinear results & Impedance mode shifts

3. **Decimate benchmarks**
   - Decimate, decimate_pro and the optimal decimation factor
   - Decimate_pro(0.80) benchmark per method for 6L2 objects

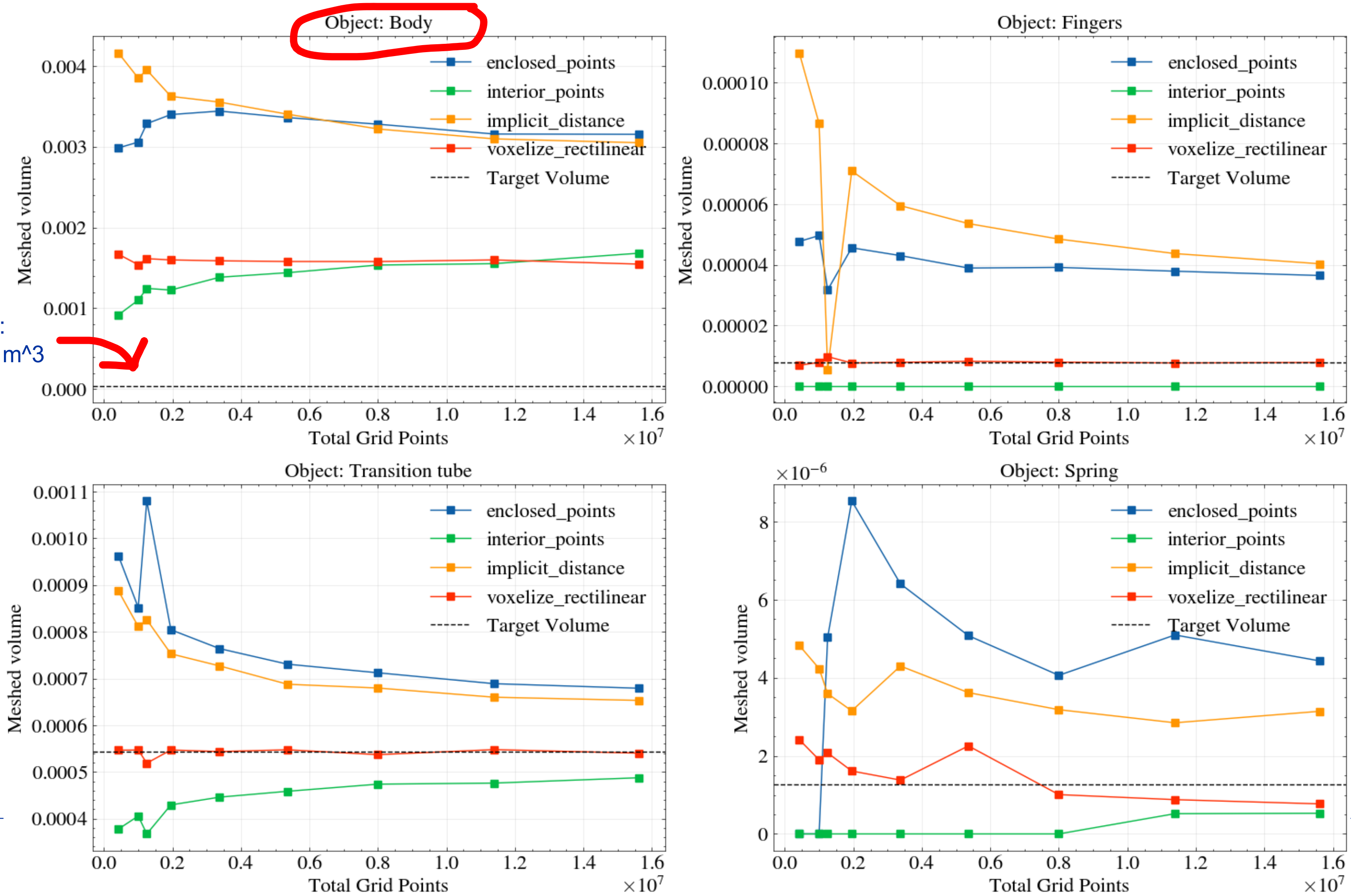
4. **The meshing algorithm times**

Benchmark shell: mask volume

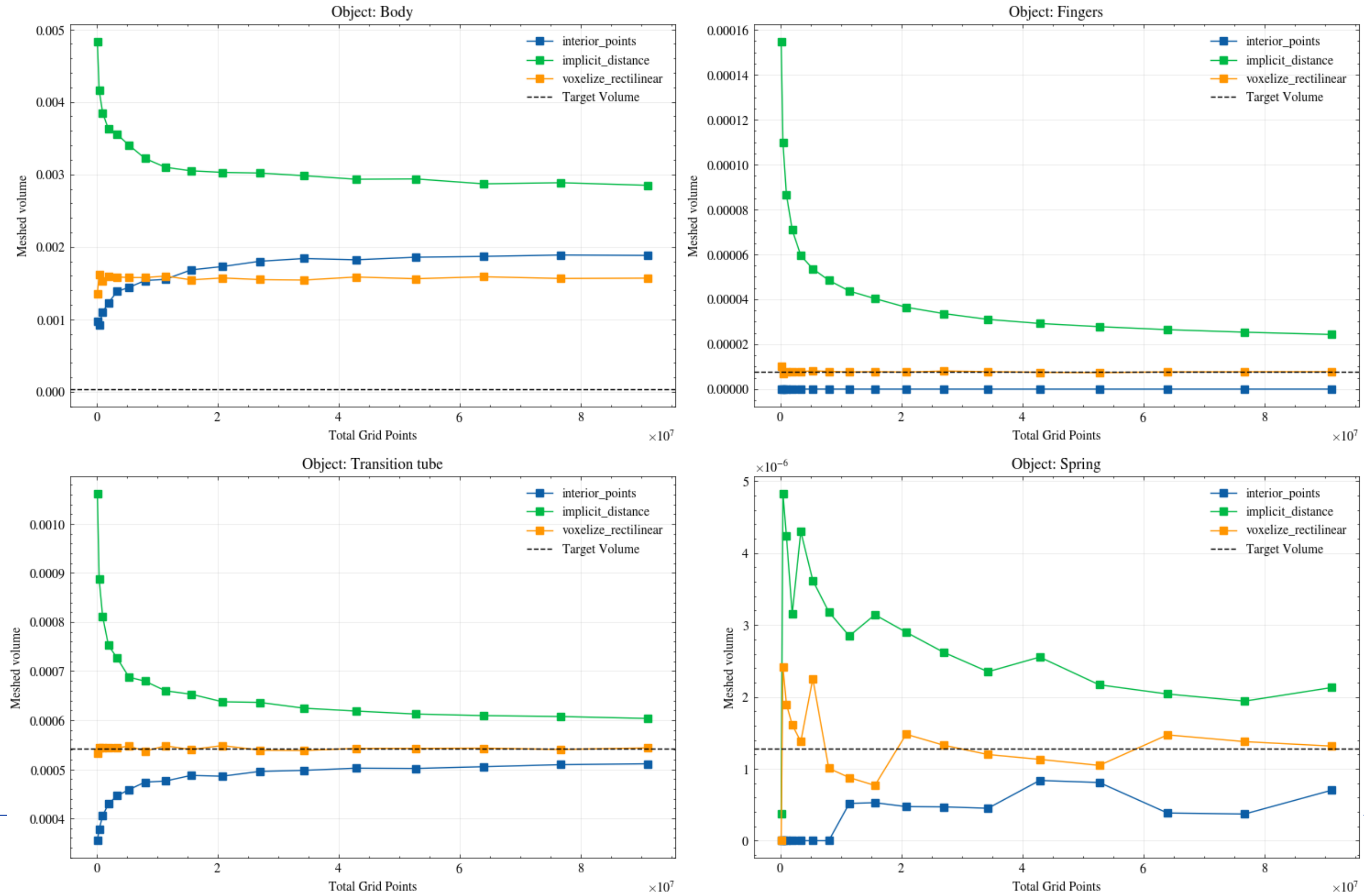


Benchmark: 6L2 sub-objects (added **N=225,250** grid resolutions)

Unphysically thin object:
Volume_body=4.3e-5 m^3



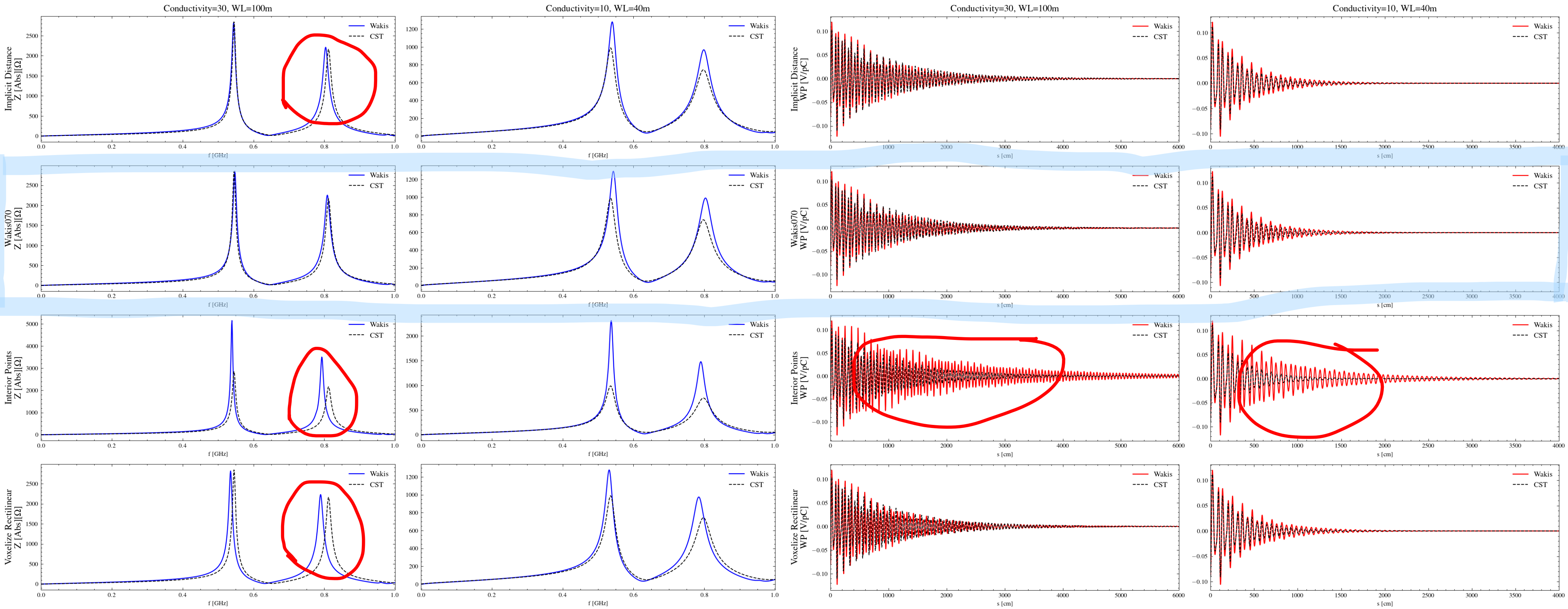
Benchmark: 6L2 sub-objects (high grid resolutions added; till **N=450 ~ 91e6 grid points**)



Simulation benchmarks to CST & the current Wakis version 0.7.0 :

The simple shell cavity

Longitudinal Impedance | **Wake potential**

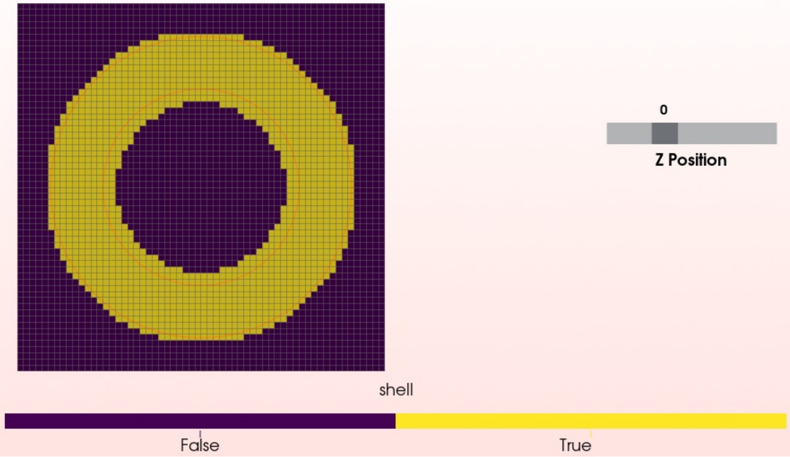


We see mode shifts for all the new methods -> new mask = new object geometry, ***inner radius***

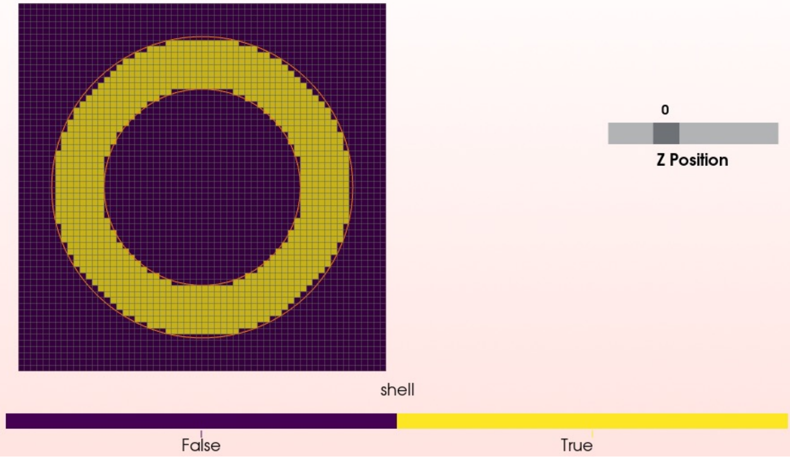
Interior_points method under-masks -> shell mask too thin -> ***conductivity***

Comparing the mask slices from these simulations

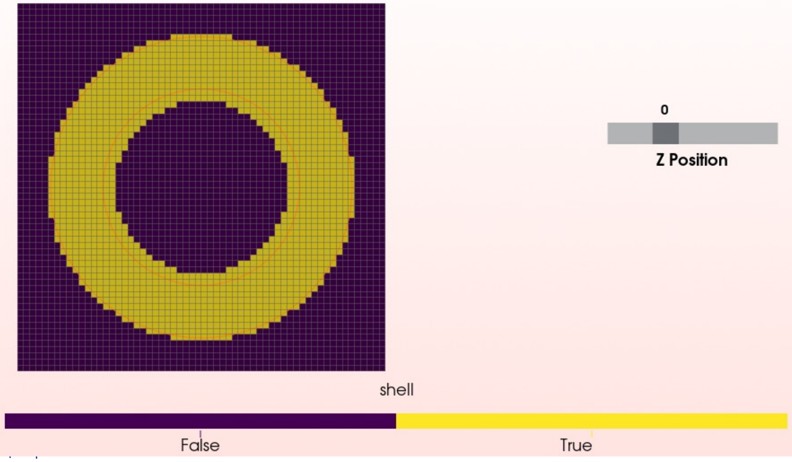
Select_enclosed_points() w. threshold=stl_tolerance



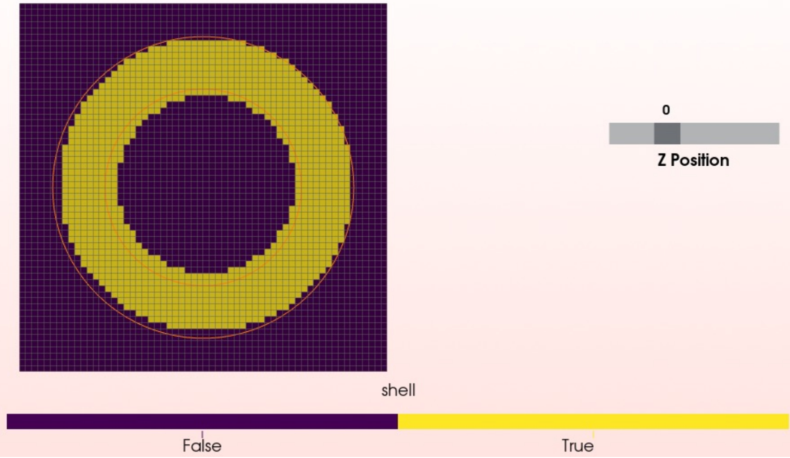
Select_interior_points() w. threshold=0.5



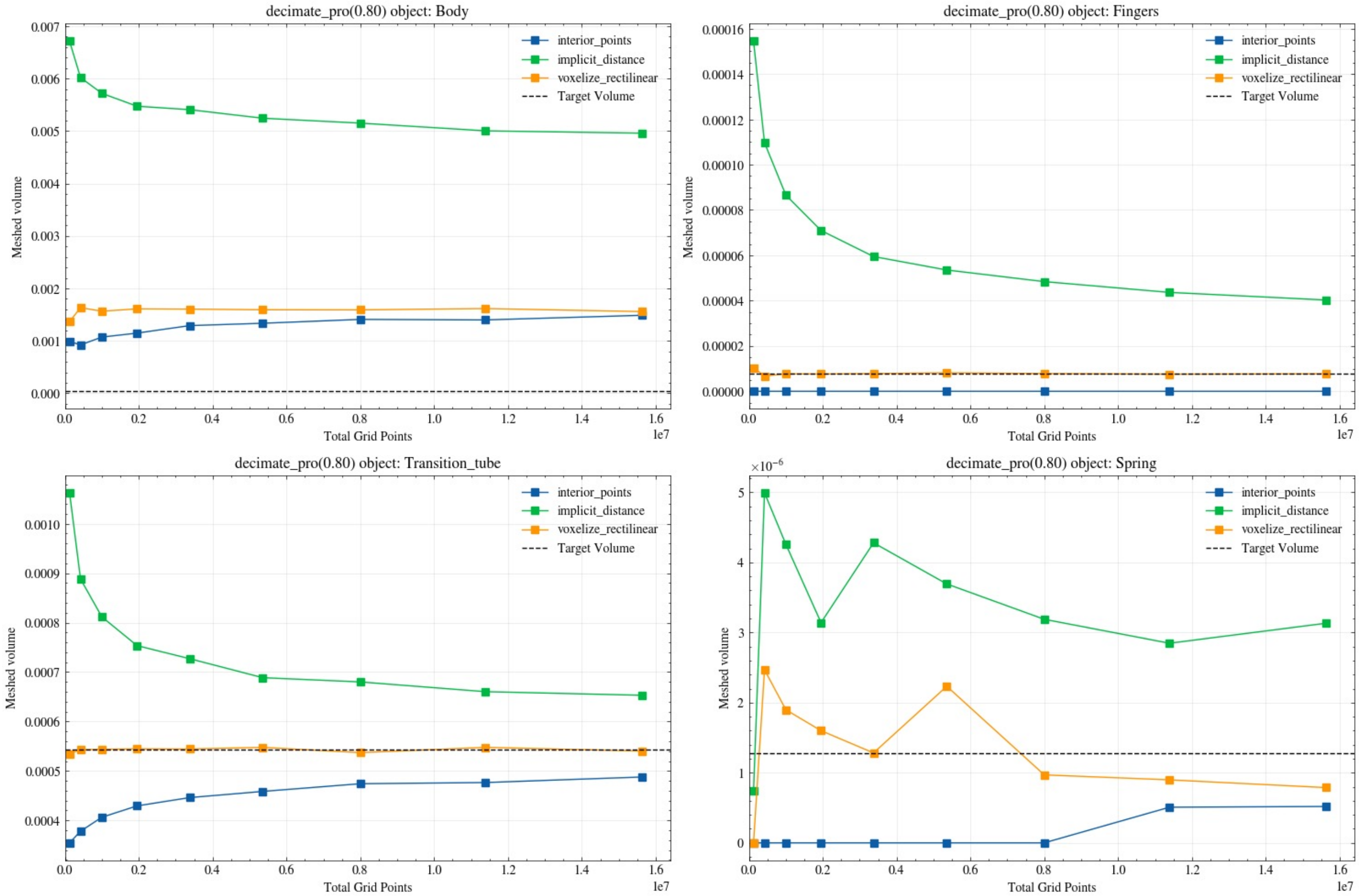
Implicit_distances()

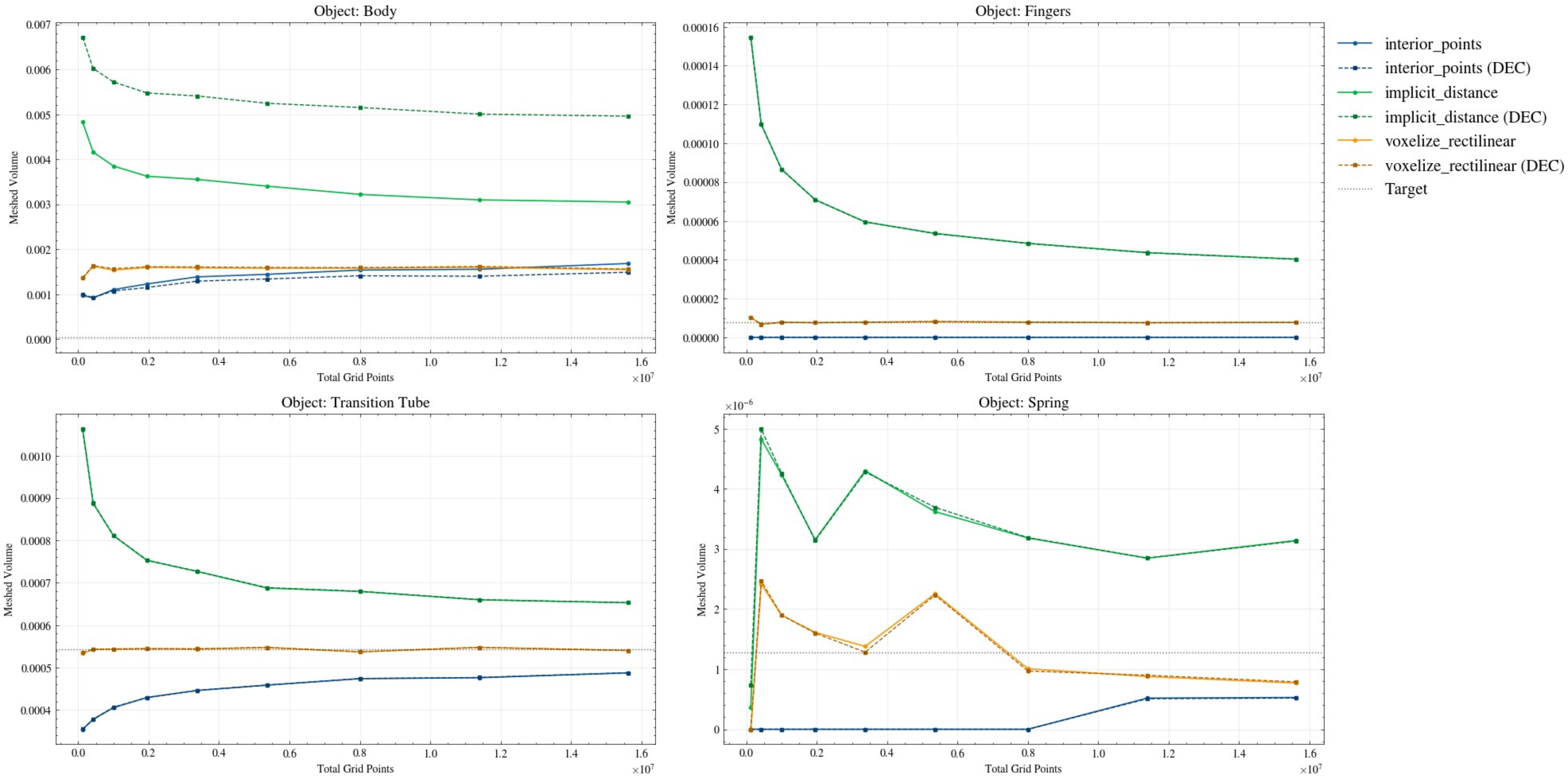


Voxelize_rectilinear()



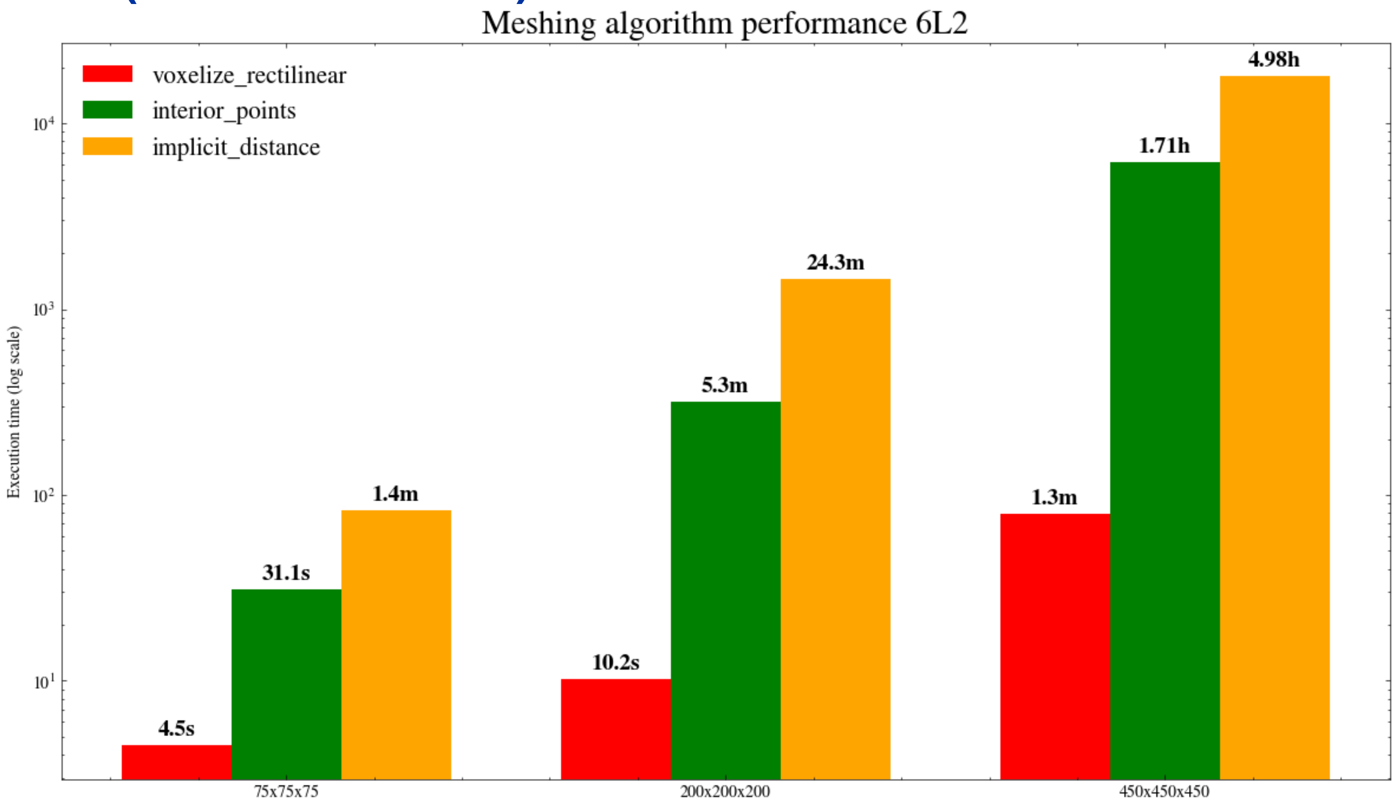
Decimate_pro(0.80): reducing 80% of STL cells





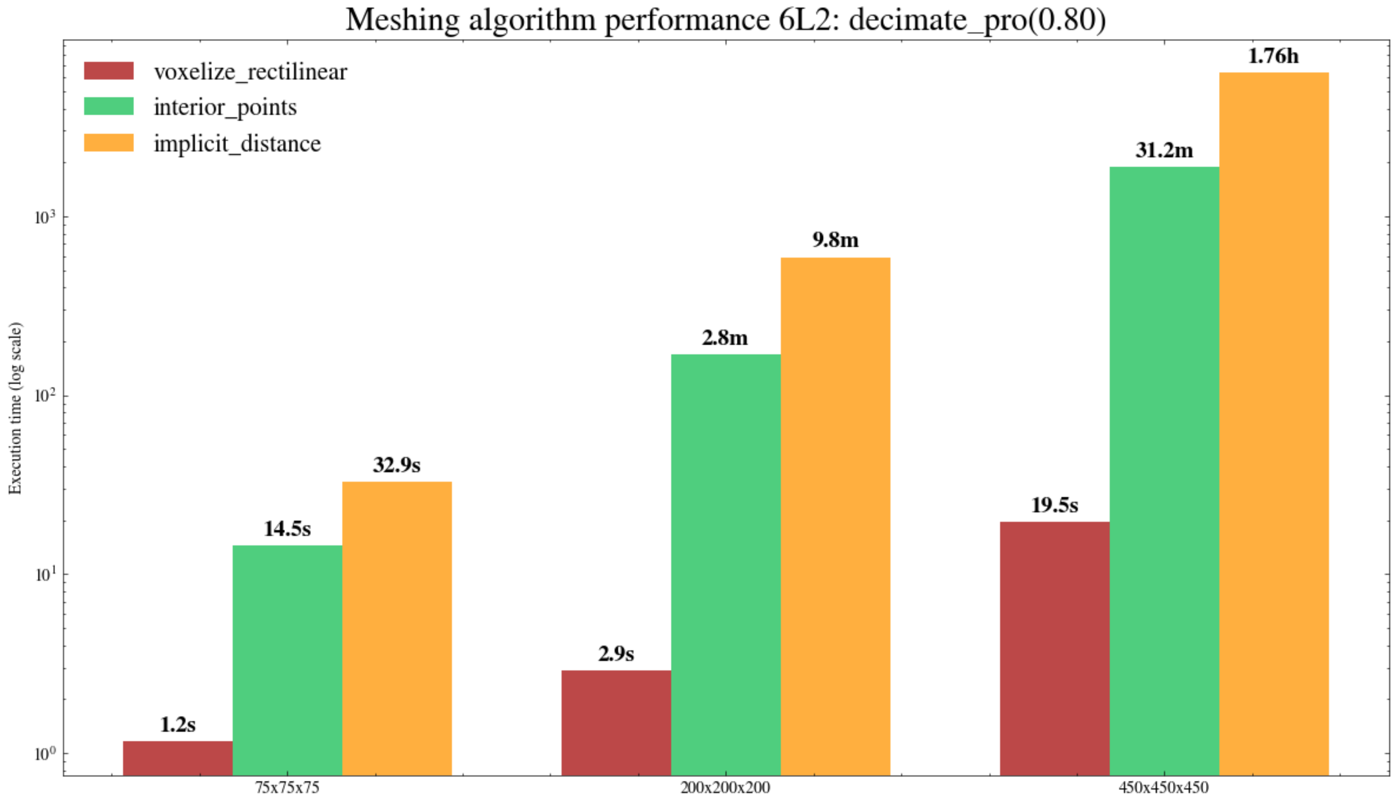
Object: Body
Object: Fingers
Object: Transition Tube
Object: Spring
Meshed Volume
Total Grid Points
×10^7
×10^-6
interior_points
interior_points (DEC)
implicit_distance
implicit_distance (DEC)
voxelize_rectilinear
voxelize_rectilinear (DEC)
Target

Scaling times of the meshing algorithms with grid resolution (& STL size)



Scaling times with 80% less STL triangles (in each of the 6L2 objects)

Taking this into account (as well as the methods' masking qualities), do we want to **proceed with** studying the **implicit_distance method ?**



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